

Project Design Phase

Problem – Solution Fit Template

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	2 Marks

Problem-Solution Fit Canvas

1. CUSTOMER SEGMENT(S) [CS] - Bank credit analysts who review applications - Compliance officers screening for high-risk applicants - Prospective customers checking their own eligibility before formally applying	6. CUSTOMER LIMITATIONS [CL] - No dedicated data-science team available to maintain a custom model long-term - Limited time per application during peak volume periods - Decisions must be explainable (not black-box) for compliance and fairness requirements	5. AVAILABLE SOLUTIONS [AS] - Fully manual review — slow and inconsistent across reviewers - Generic credit-scoring services — not tailored to this dataset or workflow - Spreadsheet rule checklists — rigid, error-prone, and hard to scale
2. JOBS-TO-BE-DONE / PROBLEMS [J&P] - Analysts need to decide approve/reject consistently across thousands of daily applications - Compliance officers need to flag applicants with 60+ days past due reliably - Customers need to know their odds before formally applying	9. PROBLEM ROOT CAUSE [RC] There is no automated, consistent way to translate an applicant's financial and credit history data into a clear, explainable approve/reject decision at scale. Raw payment-history codes (STATUS 0–5, C, X) must be converted into a binary risk label before a model can learn from them.	7. BEHAVIOUR [BE] - Analysts currently cross-check income, employment, and credit history manually against internal guidelines for every application (high effort, daily) - Judgment varies between reviewers and over time, leading to inconsistent outcomes
3. TRIGGERS TO ACT [TR] - A new credit card application arriving in the review queue - A compliance audit cycle requiring high-risk applicant identification - A customer considering applying and wanting to avoid a wasted application	10. YOUR SOLUTION [SL] Ledgerline: a Flask web application backed by the best of four trained ML classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost). It takes an applicant's financial and demographic details, engineers the same features used in training, and returns an instant, calibrated Approve/Decline decision with a confidence score — via a single-applicant form, a batch CSV screener, and an optional IBM Watson cloud endpoint.	8. CHANNELS OF BEHAVIOUR [CH] ONLINE: Web-based application form and batch CSV upload, accessible from any browser on the bank's network OFFLINE: All batch screening results are exportable as CSV for offline record-keeping and audit trails
4. EMOTIONS: BEFORE / AFTER [EM]		

Before: overwhelmed (analyst), uncertain (compliance officer), anxious (customer) After: confident in a fast, consistent decision (analyst); assured that high-risk applicants are flagged systematically (compliance); informed and in control before applying (customer)		
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